

Measuring segmentation quality without ground truth in the virtual unwrapping of Herculaneum papyri

and four artefacts the measurement reveals

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Abstract. Virtual unwrapping recovers text from carbonised scrolls by fitting a surface to the papyrus inside an X-ray tomogram and flattening it. The fitting step — **segmentation** — is the acknowledged bottleneck, and it has no ground truth: on a sealed scroll nothing says where the sheet actually is. Existing quality signals are therefore either geometric self-consistency checks or human inspection. We introduce a measurement that needs neither a threshold, nor a reference surface, nor a physical scale. It renders the same surface through progressively deeper windows and reads the exponent α of the power law relating the apparent distance-to-material to the window depth. A surface lying on its sheet gives $\alpha \approx 0$; one lying across the stack gives $\alpha \approx 1$, because its “peak” is merely the strongest thing the window happened to contain. Applying this and a complementary remote triage to 13 of the scrolls in the Vesuvius Challenge prize set, we report four findings that bear on how the community’s results are read. First, the reference tracer is **a draw, not a function**: over 78 runs with strictly identical parameters, 5 of 13 scrolls flip between a clean and a self-intersecting verdict. Second, both the **stability** and the **cleanliness** usually attributed to a good trace are artefacts of the generation budget: raising it from 120 to 400 moves the median area dispersion from 0.55 % to 86.00 %, and the share of self-intersecting runs from 1/12 to 9/12. Third, the render window imposes a second ceiling of the same kind: 13 of 16 traces sit exactly on it, so an absolute threshold on a distance compares settings rather than surfaces. Fourth, the published segmentation of a prize scroll is a set of samples — **one patch per sheet** — not a tiling of one sheet: of 105 pairs among 15 published segments, none is closer than 79 μm , twice the same-sheet threshold. We also show that a surface with $\alpha \approx 1$ is a **negative control by construction** for an ink detector, since its geometry rules out a papyrus face being within reach — the control the foundational work lacks. We argue that repetition and error bars, absent from the primary literature, are the cheapest available improvement to the field’s evidentiary standard.

1 Introduction

Eighteen hundred carbonised scrolls were recovered from a villa at Herculaneum in the eighteenth century. They cannot be opened: the papyrus is a brittle carbon foam, and every mechanical attempt has destroyed what it read. **Virtual unwrapping** replaces the physical gesture with a computational one [1] — X-ray tomography produces a volume, a surface is fitted to each sheet inside it, and the surface is flattened into a plane where ink can be detected.

The fitting step is where the difficulty lives. The community calls it **segmentation**, and its own documentation states plainly that “the problem is still unsolved. No method so far manages to perfectly fit the wanted surface to the data.”¹ The recent complete unwrapping of a sealed scroll [2] required roughly 25 hours of human work per wrap — about 775 hours in total — precisely because the automatic fit had to be inspected and corrected turn by turn.

What makes segmentation hard to improve is not only that it is hard to do, but that it is hard to **judge**. On an opened fragment one can compare a fit to a photograph. On a sealed scroll there is nothing to compare to. The consequence is visible in the literature: the complete-unwrapping paper publishes no

¹Vesuvius Challenge, scrollprize.org/unwrapping, retrieved August 2026.

tracing error rate; the spiral-fitting method [3] publishes a topological guarantee — the fitted surface is a single sheet by construction — but reports 3.20 % of **sheet traversals** anyway, because a guarantee about topology is not a guarantee about which sheet one is on.

This paper is about the judging step. We contribute:

1. **A convergence test** (Section 3) that decides whether a traced surface is on a sheet or across the stack, using no ground truth, no threshold on a physical quantity, and no scale. It compares a measurement to itself under a change of a rendering parameter.
2. **A remote triage** (Section 4) that measures three properties of a published segment from a few megabytes of its published surface volume, before any download or inference is paid for.
3. **Four empirical findings** (Section 5) obtained by applying these to 13 prize scrolls: the tracer is non-deterministic in a way that changes verdicts; the stability and cleanliness of a trace are artefacts of its generation budget; the render window imposes a second ceiling, so that an absolute threshold on a distance compares settings rather than surfaces on 14 of 16 traces; and published segments do not tile a sheet.
4. **A negative control that costs nothing extra** (Section 6.4). A surface whose convergence exponent is near 1 carries a geometric proof that no papyrus face is within reach, so any ink reported on it is a false positive by construction. This is the control the foundational paper lacks, and any pipeline that traces surfaces can produce one deliberately.
5. **An argument about method** (Section 6): every number here is a measurement repeated, and the repetition is what produced the findings. None of them would have been visible from a single run. The same move applied one step downstream — to the field’s only ground-truth-free argument about **reading** — shows that two of the three properties that argument groups together do not go together.

What this paper does not claim. It does not propose a better segmentation algorithm, and it does not read any text. It measures. Where a measurement is negative or fails to replicate, we report it as such — Section 4 contains a result that works on one corpus and fails on three others, and we publish all four.

2 Background

2.1 The data

A scroll is imaged by synchrotron X-ray tomography. The public corpus of the Vesuvius Challenge publishes, per scroll, one or more **volumes** (the tomogram, at 7.91, 8.64 or 9.362 μm per voxel depending on the scan), **predictions** (a learned per-voxel score that the voxel lies on a papyrus surface), and, for some scrolls, **segments** — the fitted surfaces themselves, together with a **surface volume**: the tomogram resampled onto a small stack of layers parallel to the fitted surface.

Ink is detected on the flattened surface. It is worth noting why the **geometry** of that surface matters so much to the reading step: the carbon ink has almost no X-ray absorption contrast, and what recent work detects is in part a **morphological** imprint — surface roughness together with pressure-induced deformation [4]. A surface fitted a few tens of microns off the sheet does not merely blur the signal; it samples a different material.

Two geometric facts matter throughout. The sheets are thin and close: on PHerc1447 we measure a median centre-to-centre spacing of 113 μm between neighbouring sheets (Section 5.5), which is about 13 voxels at 8.64 μm . And a segment is small relative to a turn: each of ours spans 8 to 22 mm of arc, between 7.7 % and 11.8 % of one revolution.

2.2 Segmentation and its tools

The reference tool of the public pipeline grows a surface from a **seed** point. Given a seed and a parameter file, it expands a quad mesh outward, generation by generation, stopping when it can no longer grow or when a **generation budget** is exhausted. The budget matters a great deal, and Section 5.2 is about that.

The tool also offers a **neighbour** mode that projects an existing surface along its own vertex normals to the next sheet, and a **resume** mode that lets an existing surface keep growing. We use both in Section 5.6.

2.3 Existing quality signals

Three kinds of signal are in use, and each has a documented limit.

Self-intersection the fitted mesh is tested for pairs of faces that cross. The official checker reports a count. It is a necessary condition, not a sufficient one: a surface can be perfectly free of self-intersections and still sit on the wrong sheet.

Topological guarantee spiral fitting [3] constructs the surface as a diffeomorphic image of a spiral, so it **cannot** be non-manifold. The same paper measures 3.20 % of sheet traversals, which shows exactly what the guarantee does not cover.

Human inspection the complete unwrapping [2] relies on it, at the cost quoted above. It is the only signal that currently catches a wrong-sheet error, and it does not scale.

What is missing is a signal that is **automatic** and **semantic** — one that answers “is this surface on a sheet?” rather than “is this surface a valid mesh?”.

3 A convergence test for a traced surface

3.1 The idea

Take a fitted surface and render it: resample the tomogram onto a stack of n layers centred on the surface. In that stack, look for the layer at which material is densest — the **peak** — and record its distance d from the surface.

If the surface sits on a sheet, the peak is that sheet, it is close, and d does not depend on n : widening the window adds layers that contain nothing nearer. If the surface sits **across** the stack, there is no sheet within reach, and the peak is simply the strongest thing the window happened to contain — so it moves outward as the window widens.

The test is therefore not a measurement of d at all. It is a measurement of how d responds to n .

3.2 Definition

Render the same surface at two depths $n_0 < n_1$ and record the two distances d_0, d_1 . Assume a power law $d \propto n^\alpha$ and read the exponent:

$$\alpha = \frac{\log(d_1/d_0)}{\log(n_1/n_0)}. \quad (1)$$

The normalisation by $\log(n_1/n_0)$ is what makes two series comparable when they were not observed over the same range: a surface looked at from 21 to 161 layers is not “worse” than one looked at from 21 to 41 merely because it was looked at further.

Two regimes:

$$\begin{aligned} \alpha \approx 0 & \iff d \text{ is a distance} && (\text{the sheet is there}) \\ \alpha \approx 1 & \iff d \text{ is the window radius} && (\text{no sheet within reach}) \end{aligned} \quad (2)$$

Why this needs no ground truth, no threshold and no scale. No ground truth, because the surface is compared only to **itself** under a change of parameter. No physical threshold, because the discriminating quantity is a dimensionless exponent, not a distance in μm — one does not have to decide how many microns is “too far”. No scale, because α is invariant under $d \rightarrow \lambda d$: it does not matter whether the voxel is 8.64 or 9.362 μm , nor whether the sheets are 113 μm apart or 300.

3.3 Behaviour on a known-good and a known-bad surface

Widening the window: what moves, and what does not

Measured distance to the nearest material, against the rendered depth. Logarithmic axes.

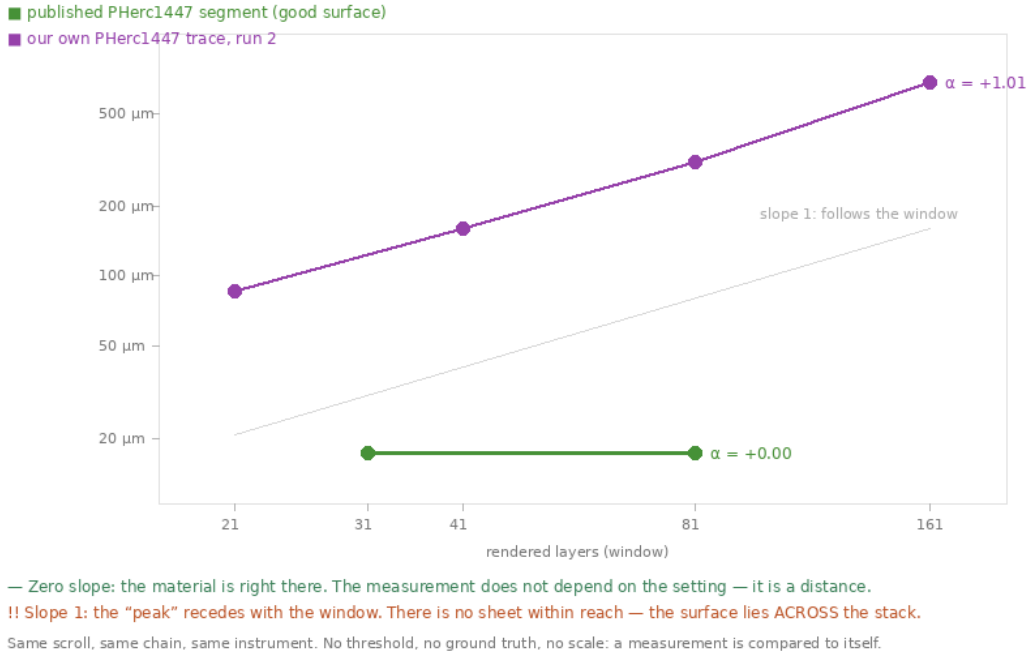


Figure 1: The test on two surfaces of the same scroll, measured with the same instrument. Both axes are logarithmic; the thin grey line is slope 1. **Green:** a published segment of PHerc1447. Its measured distance is 17.3 μm at 31 layers and 17.3 μm at 81 — a factor 1.00 for a factor 2.6 of window, $\alpha = +0.00$. **Orange:** one of our own traces of the same scroll, at 21, 41, 81 and 161 layers: 86.4, 159.8, 311.0 and 682.6 μm — a factor 7.90 for a factor 7.7 of window, $\alpha = +1.01$. The French labels are those of the original instrument; the numbers are unchanged.

The interpretation of the orange series is worth stating carefully, because it is stronger than it first appears. It is **not** “our trace is 311 μm away from its sheet”. At 161 layers the search reached 691 μm — about four inter-sheet spacings — and still found nothing. The correct reading is that **there is no sheet within reach at all**.

That the surface is lying **across the stack**, rather than merely far from everything, is settled by the rendering below and not by α alone — a distinction we make precise in Section 3.5.

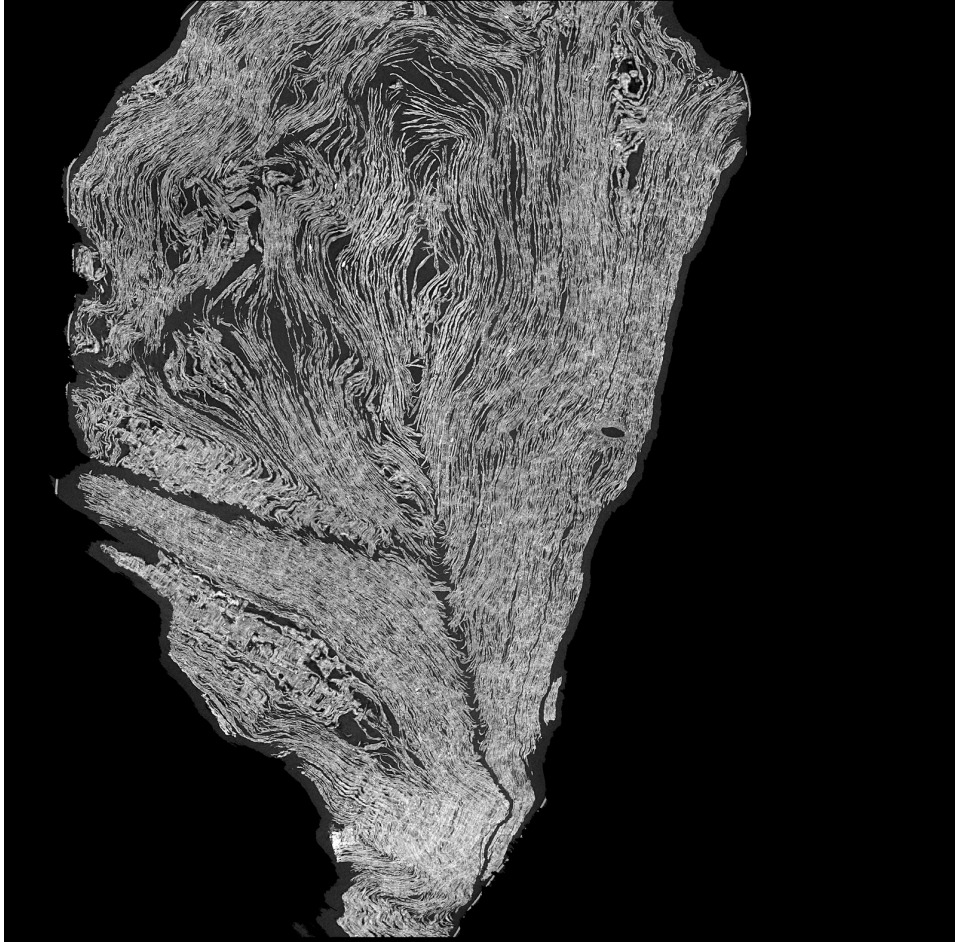


Figure 2: A rendering of the across-the-stack surface. One sees concentric laminations — the scroll seen edge-on — and not the fibrous texture of a papyrus face. This is what $\alpha = +1.01$ looks like. The convergence test puts a number on an appearance that was previously only describable.

3.4 Resolution, and a verdict that is a coin flip

An instrument must state what it cannot distinguish. On two windows, α does not discriminate better than about ± 0.2 . That has a direct consequence for how α may be used: **a count of verdicts is a count of threshold crossings**, and any verdict whose α lies within 0.2 of the decision boundary is a coin flip, not a measurement.

We enforce this in the instrument rather than in the prose. Each verdict carries its margin to the threshold and a **fragile** flag when the margin is below the resolution. Nearly a quarter of our distinct verdicts are fragile, one of them by two thousandths — and we retracted a published conclusion that rested on it the same day.

A median hides a minority. α is computed from a median over rendered windows, so a surface can converge on the median while a minority of its windows have their peak pinned to the edge of the stack — a periphery with no sheet within reach. We therefore report, alongside α , the fraction of windows whose peak is edge-pinned. A surface measured at $\alpha = +0.000$ can carry 9 % of such windows, and a chain with the **better** α (+1.040 against +1.313) can have nearly double the edge-pinned fraction (56 % against 30 %). The two measurements disagree, and both are reported.

3.5 What $\alpha \approx 1$ does not distinguish

A depth profile reports the distance to the nearest peak. When the profile is **flat** there is no peak, and the reported distance defaults to the edge of the window. The ratio of two window edges is the ratio of the two windows, so such a series yields $\alpha \approx 1$ **by arithmetic identity**, whatever the volume contains.

The two situations are different. A receding peak is a measurement: there is a sheet, it is far, and its distance tracks the window. A flat profile is not a measurement at all. Both print the same verdict.

We audited every depth profile in our tree — 225 profiles across 114 series — computing each series’ measured α alongside the α it would have if every reading were its own window edge.

Where α stops telling two failures apart

α if every reading were its own window edge

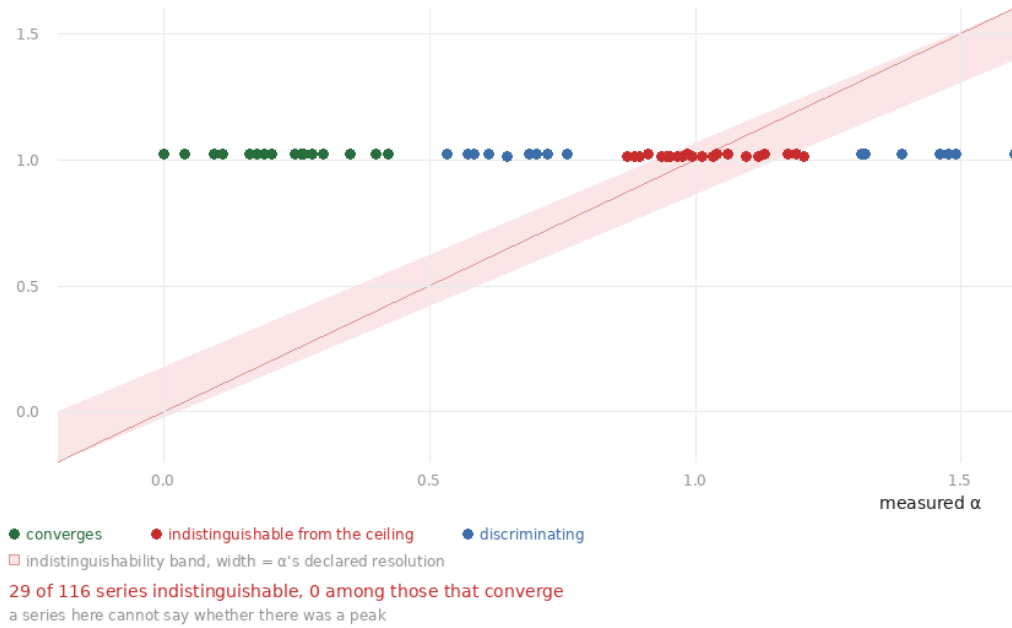


Figure 3: Each series placed on its **measured** α against the α it would have if every reading were its own window edge. The ceiling α is always near 1, so the band — whose width is α 's own declared resolution, not a display choice — collects exactly the series that cannot say whether a peak was there. The gap between the green cloud and the band is the reassurance: no converging verdict is anywhere near it.

quantity	value
profiles whose reading is the window edge	21 / 225
flat profiles (nothing to measure)	15 / 225
series where no window measures anything	4 / 114
series where α cannot separate the two failures	24 / 111
converging series so affected	0 / 111

Table 1: The audit. What loses discriminating power is exactly the **across-the-stack** verdict. The smallest non-discriminating α is +0.8729; the largest converging α is +0.4222. The two populations do not overlap.

This weakens a phrase, not a conclusion. The ceiling α is always near 1, so a converging surface is far from it by construction and no positive verdict is affected. And the two causes of $\alpha \approx 1$ — the peak recedes, or there is no peak — both mean **no sheet within the window**, which is what the verdict is used for. What over-claims is the wording “the peak moves with the window”, which presumes a peak. Our tool now reads the profile’s own amplitude against its own detection floor and returns **undecidable** rather than a confident sentence about an object that is not there.

3.6 A slope has two footings

The refusal above aggregates window amplitudes with a **maximum**: if one window has relief, the series is not empty. That rule answers “is there anything here”. It does not answer “what is the slope”, and α is a slope through two points. A footing that measures nothing is not a footing.

The remedy is not another refusal. A reading pinned to the window edge is not missing data: it is a **bound**. The peak is at least that far, possibly further. Since $\alpha = \log(e_1/e_0)/\log(n_1/n_0)$ with $n_1 > n_0$, the direction follows rather than being guessed.

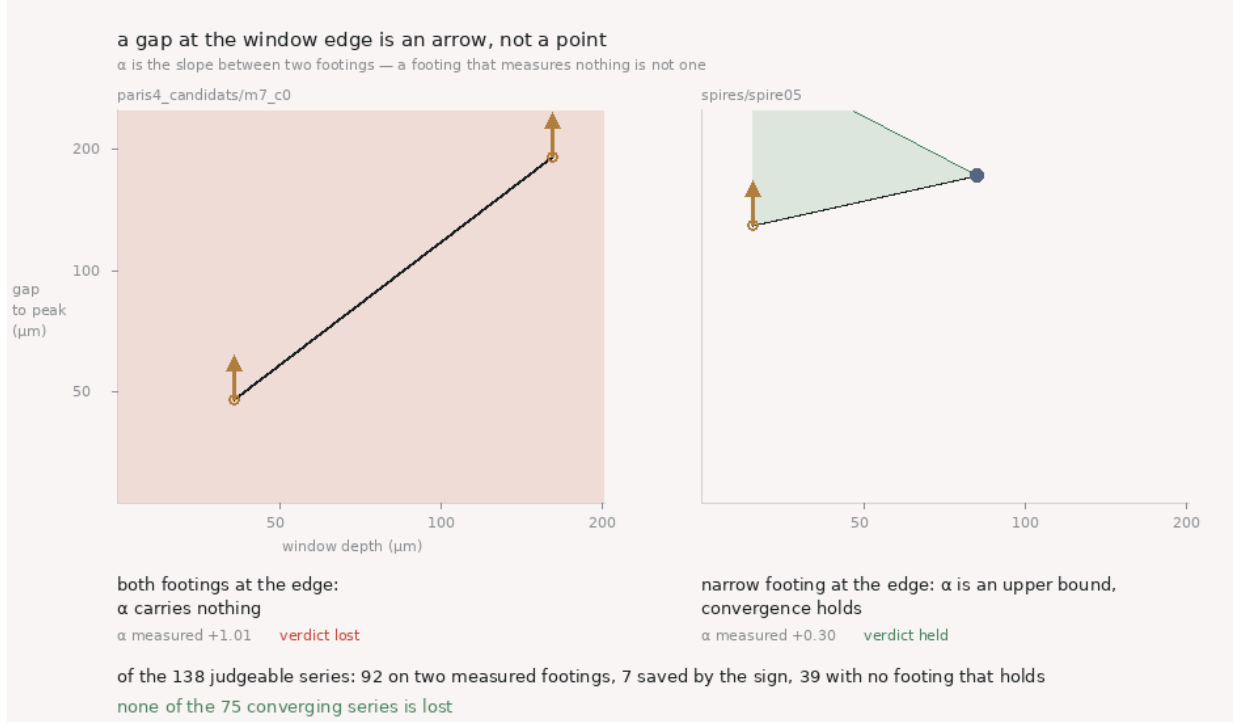


Figure 4: A gap at the window edge is an **arrow**, not a point. **Left**: both footings at the edge, so no slope is excluded and $\alpha = +1.01$ carries nothing. **Right**: only the narrow footing is at the edge, so the admissible slopes open into a single corner — all of them below the drawn line — and the converging verdict survives.

footing at the edge	α is
neither	exact
the narrow one	an upper bound: convergence holds, condemnation does not
the wide one	a lower bound: condemnation holds, convergence does not
both	nothing

Table 2: The sign of the bound decides which verdict survives. A footing that is **flat but not at the edge** gives no sign at all: its reading is the argmax of noise, neither an upper nor a lower bound.

Over the 138 judgeable series of our tree: 92 rest on two measured footings, 16 on an upper bound, **none** on a lower bound, and 30 on no footing that holds. **39 verdicts no longer stand**; 7 are saved by the sign; and **0 of the 75 converging series** is lost — because the only sign observed preserves exactly what sits below the threshold. The reassurance of Section 3.5 is thereby given a mechanism instead of an observation.

Thirty-nine verdicts that no longer stand means stacking alphas says little. But **does the narrow window show relief** is a binary question asked **inside** one window: no threshold to tune, no depths to match, no slope resting on two footings.

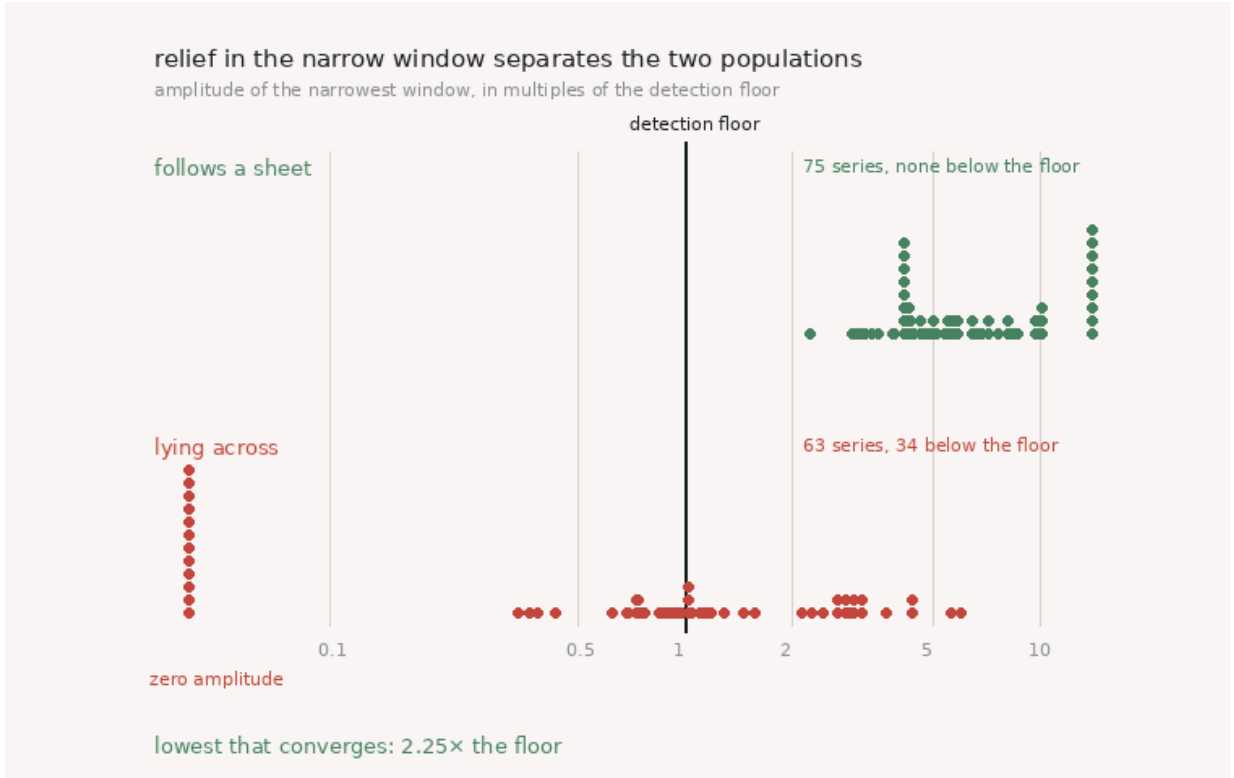


Figure 5: Amplitude of the narrowest window of each series, in multiples of the instrument’s own detection floor, on a logarithmic axis. Every series that **converges** sits to the right of the floor; the lowest is at 2.25 times it. Series lying **across the stack** straddle it, and the tall column on the left is amplitude exactly zero, which has no logarithm and is therefore placed apart rather than crushed onto the smallest non-zero value.

Not one. Zero of 75 converging series has a flat narrow window, against **34 of 63** condemned ones. On a surface that follows a sheet the relief is already there in a narrow window, which is what following a sheet means; on a surface lying across the stack it appears only by widening, that is, what one reads is the window and not the surface.

This is not circular, and the contrast is one-sided. The partition is by the sign of α , which comes from the gaps; the flatness comes from the amplitudes. Two different quantities from the same profiles, linked by a mechanism — a peak inside the window gives both a stable gap and a frank amplitude — and that link is the content of the finding rather than a flaw in it. In the other direction 29 condemned series do have a measuring narrow window: a flat one condemns, a measuring one does not absolve.

The first count was 132, and it was missing six. Two directory conventions coexist in our tree: older campaigns write the profiles of a surface side by side, while our profiling tool writes **one window per subdirectory**. Grouping by parent directory cut the latter into series of a single profile, which are unjudgeable — and they include the pyramid control and the raised-budget campaign, among the most consequential runs we have. None of the six carries a verdict. The series key is now shared and **the pyramid level is part of it**: merging two resolutions would compare two voxel sizes while appearing to compare two depths.

Twenty series returned the same number, to four decimals. When both footings are at the edge, $\alpha = \log(192/48)/\log(161/41) = 1.0135$ — a property of the **window pair** and of nothing else. Twenty independent runs, on seeds thousands of voxels apart, in two different predictions, with different draws of a stochastic tracer, return it identically. This is the lesson of Section 5.2 one level deeper: there, eight areas agreed to four significant figures because a budget truncated them all; here, twenty exponents agree to four decimals because a window pair determines them all. In both cases the **coherence** is what makes the artefact invisible, and each individual value — 0.3175 cm^2 , $+1.01$ — looks entirely plausible on its own.

4 Remote triage from published surface volumes

4.1 Why it is cheap

A published surface volume is stored as OME-Zarr with chunks of shape $[d, 128, 128]$ — one chunk is **the entire depth column of a 128×128 window**, which is exactly the unit a depth profile needs. Chunks are independent HTTP objects, so reads parallelise with no coordination. A whole segment is judged from about 300 range requests, a few megabytes and some fifteen seconds; we measure a speed-up of $\times 8.35$ at 16 threads with byte-identical output.

4.2 What is measured

field	meaning	evidence
material	fraction of probed windows containing papyrus	$\rho = +0.539$
edge_pinned	peak sits at a stack edge — the sheet is outside the volume	$\rho = -0.275$
offset	median distance from the traced layer to the material peak	$\rho = +0.388$
residual	what remains after the best rigid shift	$\rho = +0.428$
rigid_share	share of the error a mesh translation would remove	21.7 %
coherence	does a window’s error predict its neighbour’s	150/152

Table 3: Fields measured remotely, and their correlation with an independent target: the published ink maps of 80 `Scroll 1` segments, taken as-is. Nothing of ours enters the target, so the relationship cannot be a shared artefact. `coherence` is compared to each segment’s own shuffle control.

4.3 A rule, and its failure to replicate

On `Scroll 1`, dropping the 20 % of segments poorest in `material` raises the corpus median ink contrast by +0.381 against 2000 random draws of the same size ($p = 0.0005$). The threshold sits on a plateau — 15 % to 25 % all hold at $p \leq 0.001$ — which is what an honest knob looks like; an overfitted one produces a peak.

We then tested the same measurement on 110 further segments across three more scrolls, all with published surface volumes and published ink maps:

corpus	n	voxel	target spread	ρ
detectable ρ				
<code>Scroll 1</code>	80	2.4 μm	1.008	+0.539
0.309	<code>PHerc0139</code>	38	2.399 μm	1.628
-0.229	0.441	<code>PHerc1667</code>	19	2.399 μm
0.720	+0.425	0.605	<code>PHerc0172</code>	53
7.91 μm	0.220	-0.217	0.377	

Table 4: The rule does not replicate — and the last column says why that sentence must be read carefully. **Detectable ρ** is the correlation each corpus could find at 80 % power and $\alpha = 0.05$, given its n alone. Only `Scroll 1` exceeds its own floor. Our first explanation was a target floor effect — `PHerc0172`’s ink maps barely differ from one another — but that explains one corpus out of three: `PHerc0139` sits at the same resolution as `Scroll 1`, has a **larger** relative spread, and returns the opposite sign.

The honest reading is narrower than “it does not replicate”. **None of the three had the power to detect an effect the size of the one measured on `Scroll 1`**: each $|\rho|$ falls below its own 80 %-power floor. `PHerc1667` in particular returns +0.425 — the same sign and a comparable magnitude — on 19 segments, where 0.605 would be needed. Two corpora do return the opposite sign, which no amount of power explains away, and that is the part of the shape that suggests a result which will dissolve.

We therefore report the rule as a property of Scroll 1’s published corpus, **not** of the problem, and we publish all four corpora rather than the one that works: a reader learns at once that it needs re-validating on their corpus, instead of discovering it afterwards.

A zero is reported with its power, or it is not reported. Writing “three non-positive” — as an earlier draft of this section did — puts a corpus returning $+0.425$ in the same bin as one returning -0.229 , and hides that neither could have seen the effect. The three negatives constrain the rule far less than their count suggests.

5 Results

5.1 The tracer is a draw, not a function

The public tracer is multi-threaded and draws from an unseeded generator. Running it twice with **strictly identical** parameters, seed and input does not return the same surface. This is not documented, and no repetition or error bar appears in the primary literature.

We measured it as a paired campaign: 13 prize scrolls, six runs each, 78 runs, every parameter fixed.

The same call, six times — across thirteen scrolls

Same seed, same parameters. If the tracer were deterministic, every row would be a single point. 13/13 are not.

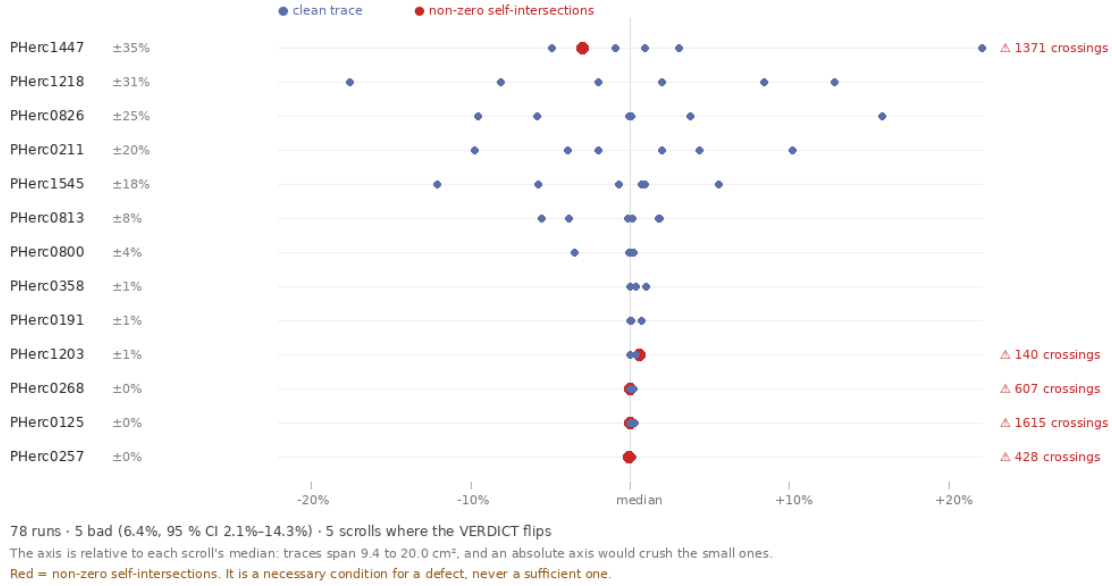


Figure 6: Thirteen scrolls, six runs each, identical parameters. If the tracer were deterministic, every row would be a single point. None is. On five rows one point is red: the same call that returns a clean trace five times returns a self-intersecting one the sixth.

quantity	value
runs, scrolls	78 over 13
scrolls whose six runs return the same area	0 of 13
scrolls where the verdict flips between runs	5 of 13
bad runs	5 / 78 = 6.4 % (exact 95 % CI 2.1 %–14.3 %)

Table 5: Non-determinism of the reference tracer, measured.

A verdict flip is the strongest evidence available here, because it needs no threshold and no ground truth: two executions with identical inputs, two opposite answers from the official checker. The five flips carry 1615, 428, 607, 140 and 1371 self-intersections respectively, against zero on the other five runs of the same scroll.

And the area does not signal the bad run. If a bad run could be recognised by its area being unusual, one could discard it without judging it. It cannot: over the five bad runs the mean centre-normalised rank of the area is 0.68, where 0.5 is expected if the area says nothing. One has to judge the trace, not measure it.

5.2 Stability and cleanliness are artefacts of the generation budget

In the campaign above, the scrolls with the lowest area dispersion are exactly those whose six runs stop at the same generation count. Two readings compete: either those scrolls are genuinely more stable, or their dispersion is **crushed by a common truncation** — six traces cut at the same place necessarily have the same area.

The measurement that separates them is cheap: raise the budget and replay. We did, on two of them, changing nothing else.

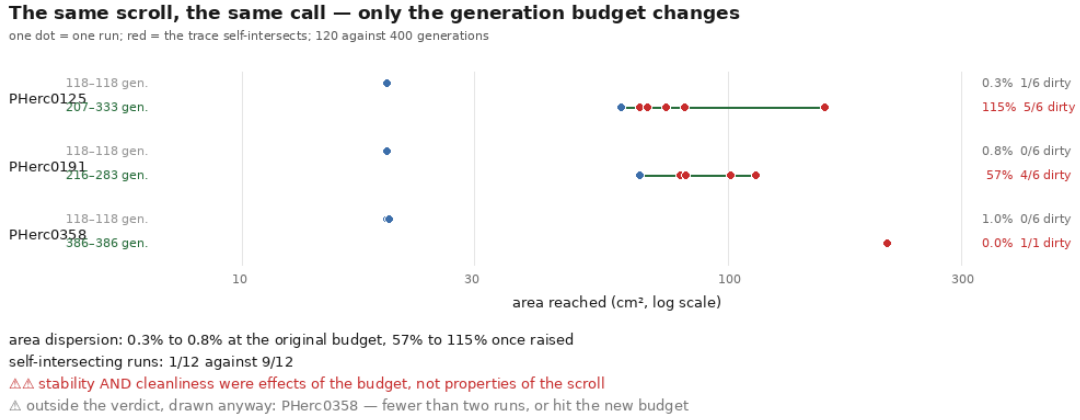


Figure 7: The same scroll, the same call, six runs each; only the generation budget changes (120 against 400). One dot per run, red if the trace self-intersects. The upper row of each pair is the original budget — six runs collapsed onto one point — and the lower row is the raised budget.

scroll / budget	generations	median area	dispersion	dirty runs
PHerc0125 / 120	118–118	19.83 cm ²	0.3 %	1/6
PHerc0125 / 400	207–333	71.31 cm²	115 %	5/6
PHerc0191 / 120	118–118	19.83 cm ²	0.8 %	0/6
PHerc0191 / 400	216–283	80.39 cm²	57 %	4/6

Table 6: Raising the generation budget. Median dispersion over the two scrolls moves from **0.55 %** to **86.00 %**, a factor 156.

Two conclusions follow, and the second was not what the experiment was designed to find.

The stability was a truncation. Six traces cut at the same generation have the same area by construction. It was not a property of the scroll; it was the budget cutting them.

The cleanliness was a truncation too. Self-intersecting runs go from 1/12 to 9/12. PHerc0191 did not flip at all at the original budget — zero dirty runs out of six — and flips 4 times out of 6 once released. **The traces were clean because they were short.**

Scope, and cost. Two scrolls entered the verdict, not thirteen. A third, PHerc0358, goes the same way — 19.83 to 211.43 cm², a factor 10.7, with 9907 self-intersections — but contributes only one run, and a single number does not disperse; including it would lower the median for a reason unrelated to the phenomenon, so our tooling excludes it **by name**. The campaign was stopped after 13 runs of 24 because at the raised budget a single run takes 40 minutes against 2 to 4, the fringe growing with the generation count. Anyone tempted to “just raise the budget” should price that first.

The same artefact contaminates a comparison we had published. Two seed criteria were compared on the same 13 scrolls, paired, one seed per criterion, everything else identical. The sign test on the reached area gives 11–2 in favour of the planarity criterion, $p = 0.0225$. But 7 of the 13 planarity traces stop at the budget — and the two “defeats” are exactly the two scrolls on which **both** criteria stop at the budget. Their area differences, 0.09 and 0.001 cm^2 , do not compare two criteria; they compare two truncations at the same place.

Paired campaign of seed criteria — 13 prize scrolls

same scroll, same volume, same parameters; only the seed changes

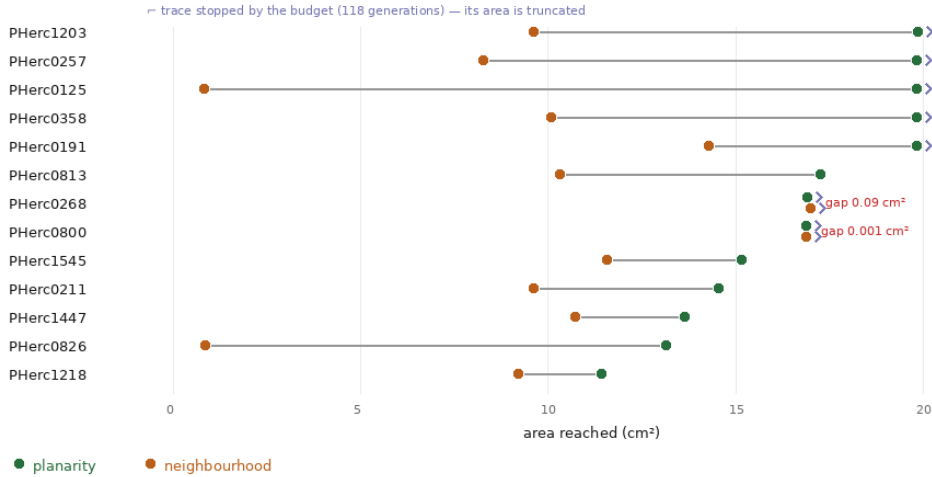


Figure 8: The paired seed campaign. Each row is one scroll traced twice with everything held fixed but the seed. A chevron marks a trace stopped by the budget: on PHerc0268 and PHerc0800 both traces carry one. On the 11 informative pairs the sign test is 11–0, $p = 0.0010$.

We keep the conservative number. The headline stays 11 – 2, $p = 0.0225$, over all 13 pairs. Publishing only the more favourable figure once one has seen which is more favourable is choosing one’s sample after the fact. The two excluded pairs are excluded by a rule stated before looking at what excluding them does.

And we then failed to apply it to ourselves. Every trace we made on PHercParis4 — the sixteen cells of the prediction cross, the eight seed candidates of Section 5.3 — was run at 60 generations, and all of them stop at generation 59. Their areas agree to **0.06 %** (0.3174 to 0.3176 cm^2) across seeds separated by thousands of voxels in two different predictions. That agreement is exactly the signature this section teaches one to recognise, and we read it as nothing at all for a week. The budget itself was not a choice: it was set when we estimated render throughput at 57 KiB/s from a **single** observation, later measured at 1108–5861 KiB/s.

The general lesson is worse than the local one. A setting adopted for a reason that has since stopped being true never announces itself; the **coherence** of the results it produces is precisely what makes it invisible. Eight traces agreeing to four significant figures look like a robust measurement.

5.3 Seed choice does not predict convergence

Section 4 left one lever untried. Our seed finder ranks candidates by **planarity alone**, and on one prediction the seed every earlier trace had used scored 1.0000 on **nine** supporting neighbours — less supported than 0.987 on **twenty-seven**. Three ten-thousandths separate the planarities while occupancy varies by a factor of thirty-five.

We did not invent a composite ranking to break the tie: choosing the weights is choosing the answer before measuring it. We traced all eight candidates.

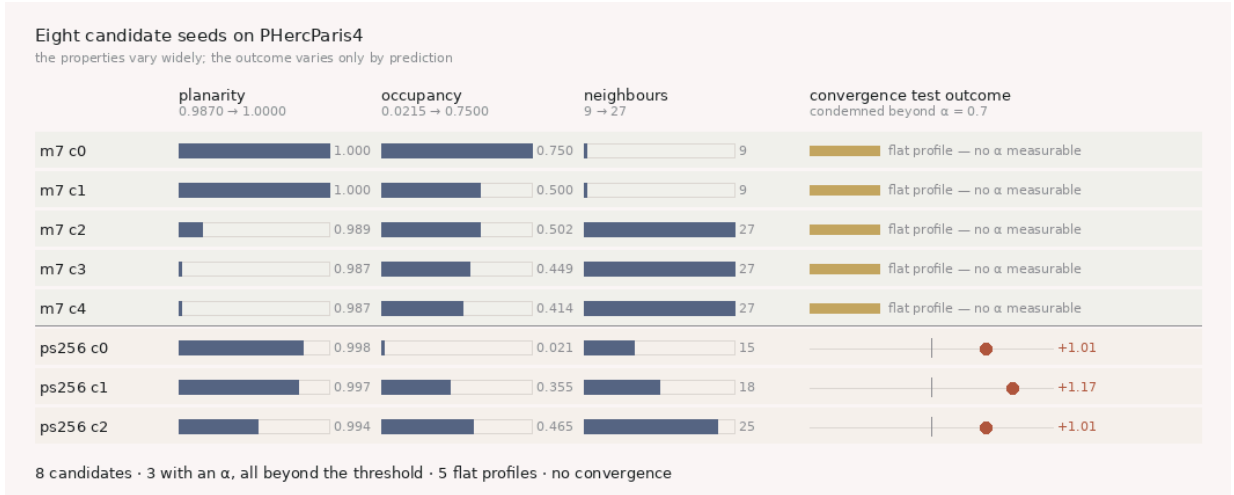


Figure 9: Eight seed candidates on PHercParis4. Each property is drawn on **its own observed range**, printed above the column — a full planarity bar spans 0.987 to 1.000, not 0 to

1. Five candidates return a profile too flat to measure at all; three return

$\alpha \approx 1$ — and Section 3.6 shows that those three do not measure either. The properties vary widely; the outcome does not vary at all.

Zero of eight converge. The lowest α obtained is +1.01 against a condemnation threshold of 0.7. The five candidates of the second prediction all fall under the flat-profile refusal of Section 3.5: they report window **edges**, whose ratio is the ratio of the windows, so $\alpha \approx 1$ by arithmetic identity whatever the volume contains.

And zero of eight measure. The remaining three, which we first reported as yielding an α , have their narrow window **below the instrument’s detection floor**; two of them have **both** readings pinned to the half-window. By Section 3.6 none of the eight rests on two footings that carry a verdict. Of the 27 PHercParis4 series in our tree, **26 fall and one holds**: ps256_sur_graine_ps256, at $\alpha = +1.12$ on two measured footings, still condemning the trace. The wall stands on one measurement rather than twenty-seven — harder to state, and easier to defend, since twenty-seven numbers of which twenty are the same arithmetic identity are not twenty-seven witnesses.

We decline to report a correlation, and that is the point. Only three candidates yielded an α at all — and by Section 3.6, none of the three carries information. With eight points, the correlation detectable at 80 % power and $\alpha = 0.05$ exceeds **0.84**: nothing short of a near-perfect relationship would be visible. A middling ρ here would not be an absence of effect, it would be an absence of power, and reporting it as the former is the most common way a null result is manufactured.

5.4 A second ceiling: the render depth

The budget of Section 5.2 is not the only one. Rendering a surface flattens a window of n layers around it, so a measurement of **how far the nearest matter lies** cannot report a distance larger than that window. It reports the window.

We rendered the same 16 traces, across 4 scrolls, at two depths — 21 and 41 layers — and paired them by identity rather than by list position.

A criterion measured at one depth does not judge a trace rendered at another

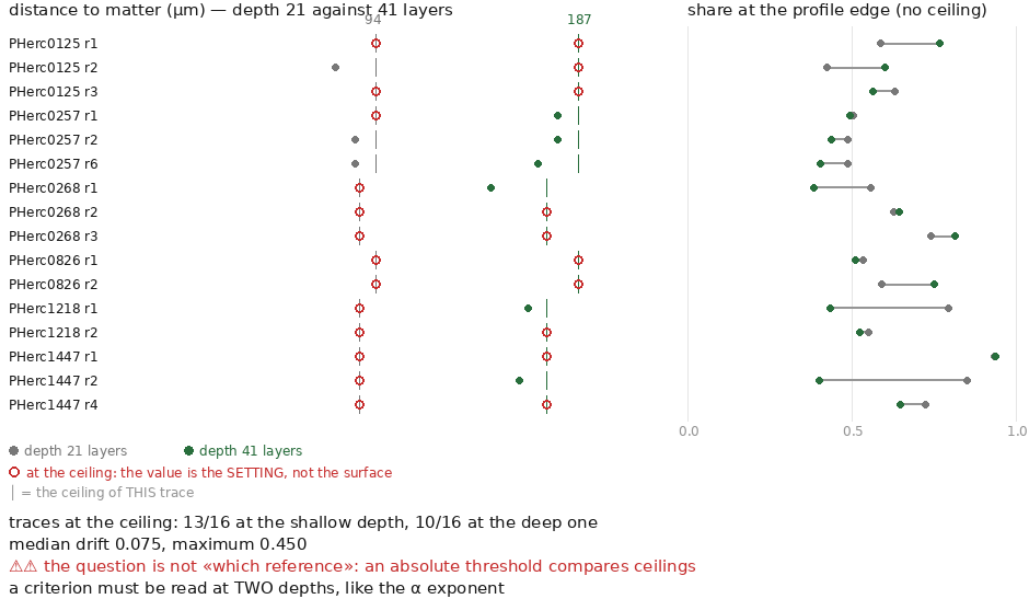


Figure 10: Left: each trace’s distance to matter at both render depths, with the ceiling of **that trace** drawn as a vertical tick. Hollow red marks sit exactly on their ceiling: the value is the setting, not the surface. Right, on the **same rows**: a criterion with no ceiling — a bounded fraction — moving between the same two depths. A trace pinned to its ceiling on the left can still travel a long way on the right.

criterion	compared	censored	median drift	max drift
distance to matter (μm)	2	14	88.939	93.620
share at profile edge	16	0	0.075	0.450
share of flat columns	16	0	0.153	0.250

Table 7: The same 16 traces at render depth 21 and 41. The ceiling doubles with the depth ($93.62 \rightarrow 187.24 \mu\text{m}$, a factor 2.00 for a factor 1.95 of depth), and 13 of 16 traces sit on it at the shallower depth, 10 of 16 at the deeper one.

Two things follow. First, a threshold on the distance compares **settings** rather than surfaces on 14 of 16 traces — the same failure as the generation budget, in a different part of the pipeline. Second, a criterion that **cannot** be truncated drifts anyway: a median 0.075 on a quantity whose typical value is 0.5, and a maximum of 0.450, which is almost the full range of the criterion, produced by changing nothing but the render depth.

The cohort has two ceilings, not one. It mixes two voxel sizes, 8.64 and 9.362 μm , so the ceiling is proportional to the scroll’s voxel: 86.40 and 93.62 μm at the shallow depth. Drawing a single cohort-level ceiling makes the traces of the other voxel look as though they sit **below** the ceiling when they sit exactly **on** theirs, and half the censoring disappears from the figure without a single number changing. Each trace carries its own tick above.

The consequence for practice is not that a better reference must be found. It is that an absolute threshold on such a criterion is not a statement about a surface. The convergence test of Section 3 escapes this by construction: it is an **exponent** relating two window sizes, so it has no reference, no threshold and no scale. A profile-based criterion read the same way — at two depths, as a ratio — would inherit the same property. We have not built it: it needs traces uncensored at both depths, and this corpus contains two.

5.5 Where a radial chain actually sits

The neighbour mode projects a surface onto the next sheet, and chaining it produces a sequence of surfaces that the convergence test does not condemn — the first such sequence we obtained. It is tempting to read that sequence as a piece of unrolled papyrus. It is not.

To locate a surface inside a scroll one would normally need the scroll's axis, which is unknown. The following observation removes the need. On a cylinder, a grid line running along the circumference **curves**; a line running along the axis is **straight**. One does not need to know where the axis is to know which grid direction is which — only which one bends.

Curvature is read from the **sagitta** of the arc, the maximum distance between the arc and its chord. For an arc of half-angle φ , the chord is $2R \sin \varphi$ and the sagitta $R(1 - \cos \varphi)$, so

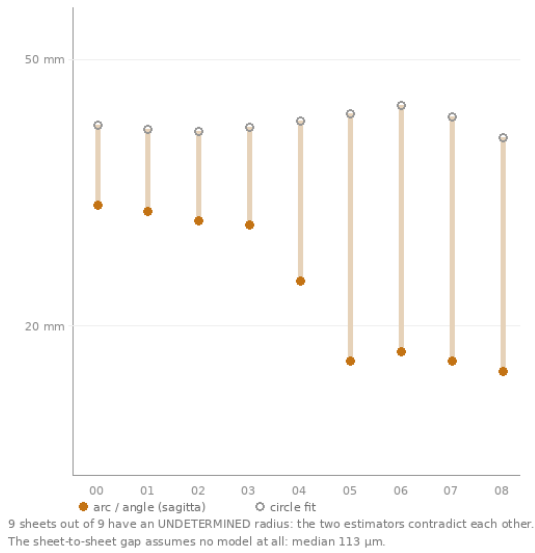
$$\varphi = 2 \arctan\left(\frac{2s}{c}\right), \quad (3)$$

with s the sagitta and c the chord, exactly and with no small-angle approximation. The radius cancels, which is what makes the quantity usable where the radius itself is not determined.

Where the chain sits inside the scroll

9 sheets — what is measurable, and what is not

Sheet radius — two independent estimators



One unrolled turn — what the chain covers

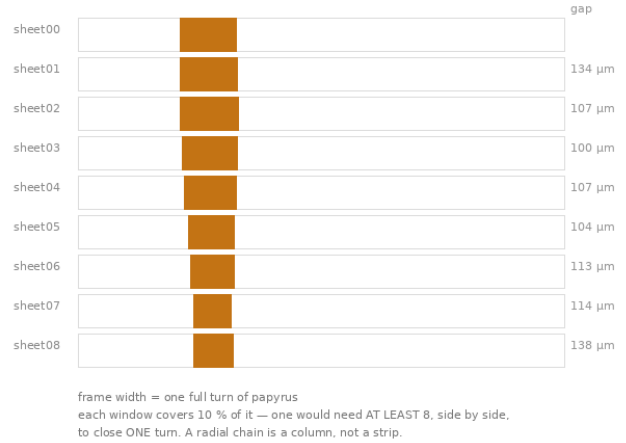


Figure 11: Nine consecutive surfaces of a radial chain, located inside the scroll without knowing its axis. Each covers between 7.7 % and 11.8 % of one revolution, and consecutive surfaces sit a median 113 μm apart — one sheet.

The measurement gives the median gap between consecutive surfaces as 113 μm (range 100–138), which is one sheet: the chain advances one sheet at a time, as designed. But each surface covers only about a tenth of a turn, and consecutive surfaces occupy **the same angular window**. Along the papyrus they are separated by a full circumference that we do not possess.

A radial chain is a **column**, not a **strip**. Gluing its members end to end would produce a band of papyrus that does not exist. Reaching a **length** of unrolled text requires a tangential chain.

Erosion bounds such a chain before quality does. On the surface actually carrying material — not the grid — it is 15.6 % per turn, so half is gone in four turns; the fraction of valid vertices falls from 58 % to 23 % along nine wraps.

5.6 Growing sideways, and a fixed point

The natural response is a **tangential** extension: let a published segment keep growing along itself. It works, once.

surface	useful area	arc	valid vertices	α
published segment, as downloaded	4.28 cm ²	21.9 mm	59 %	+0.000
after one extension	12.97 cm²	37.2 mm	96 %	+0.000

Table 8: One tangential extension of a converging published segment. Three times the useful area and 70 % more arc, and the convergence test still does not condemn it.



Figure 12: The extended surface, rendered. The material is continuous across the extension; the periphery is where the newest — and worst — vertices live.

The first time we measured this, it was a coin flip, and saying so is the point. Three runs of identical parameters gave 0, 596 and 0 self-intersections, and α of +0.000, +0.422, +0.000. Pinning the generator’s seed and forcing a single thread makes it reproducible — two runs then give the same mesh, bit for bit — and every figure above is from the pinned configuration. A result that replicates two times out of three is not a result, and it took a repetition to see it.

More growth does not give more surface. At budget 200 the area reaches 28.62 cm² with 25 036 self-intersections and $\alpha = +1.313$; split into two sessions of 100 it reaches 50.30 cm² with 4996 crossings and $\alpha = +1.040$ — better, and still across the stack. What decides the outcome is not the size of the step but **how clean the surface it starts from is**, measured on its periphery.

That gives a repair. Every vertex carries the generation at which it was created, so trimming the late periphery is a filter, not a guess. Trimming to the first ten generations returns 6.02 cm² with a periphery as clean as the source’s — 41 % more validated material than the published segment, at $\alpha = +0.000$.

And then the cycle closes on itself. Extending that clean surface again also converges, but trimming the result back to a clean periphery returns 6.02 cm² again, exactly. The trim-and-extend cycle does not diverge; it **converges to a fixed point near 6 cm²**. Each turn regains what it just gave up.

We report this because a cycle that appears to work for one iteration is exactly what a reader would extrapolate from, and it does not.

5.7 Published segments do not tile a sheet

The remaining route to a continuous strip is to stitch published segments together. The tool exists and needs surfaces that **overlap**.

PHerc1447 publishes 15 segments. Of their 105 pairs, 51 overlap by bounding box, many by 90–100 %. But a box overlap cannot tell “two patches of one sheet” from “two adjacent sheets”, which in a scroll occupy nearly the same volume. The discriminant is the median point-to-point gap.

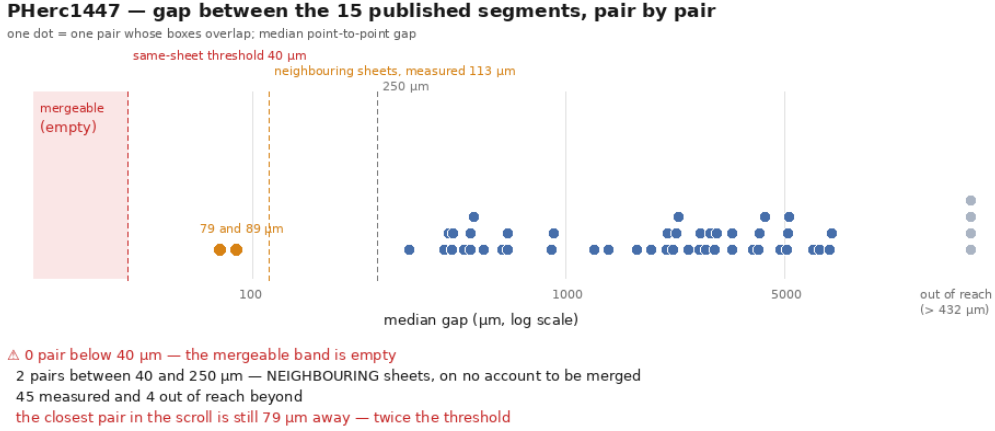


Figure 13: Every overlapping pair of published segments, placed on its median point-to-point gap (logarithmic axis). The band below the same-sheet threshold is **empty**, and the closest pair in the whole scroll is still 79 μm away — twice the threshold, and of the order of the measured inter-sheet spacing.

median gap	meaning	pairs
< 40 μm	two patches of the same sheet — mergeable	0
40–250 μm	neighbouring sheets — must not be merged	2
≥ 318 μm	several sheets apart	45 measured, 4 out of reach

Table 9: Pairwise gaps among 15 published segments of PHerc1447. Establishing this cost 4.5 MB of download and a few seconds.

The published segmentation of this scroll is a set of samples — **one patch per sheet** — not a tiling of one sheet. There is nothing to stitch. Combined with the fixed point of Section 5.6, both named routes to a continuous strip are measured and closed, each for its own reason.

6 Discussion

6.1 Repetition is the cheapest available improvement

Every finding in Section 5 came from doing the same thing twice. The tracer’s non-determinism is invisible in one run. The truncation artefact is invisible without a second budget. The coin-flip nature of the tangential extension is invisible without a third repetition. None required a new algorithm, a new dataset, or a new model.

The primary literature publishes neither repetitions nor error bars for the segmentation step. Given a tool that returns a different surface on every call, a single reported surface is a sample from a distribution whose width is not reported — and we measure that width to be, in the extreme, 115 % of the mean.

6.2 The truncation artefact generalises beyond this pipeline

The mechanism is not specific to one tool. Any iterative fitting procedure with a budget will, on the instances where the budget binds, produce outputs that agree with each other for a reason that has nothing to do with the instance. Reporting a dispersion over such outputs measures the budget. The diagnostic is cheap: record the stopping condition alongside the result, and check whether the low-variance instances are the ones that hit it.

We met the same mechanism three times in this work, in three unrelated parts of the pipeline: the generation budget of the tracer (Section 5.2), the render window of the flattening step (Section 5.4),

and — in the opposite direction — the fact that the **absence** of a stopping condition is what made the convergence test of Section 3 transportable. A quantity that a setting can cap is a quantity whose reported spread is, in part, a property of the setting.

6.3 The same move works one step downstream

The instruments above judge **geometry**. The field’s only ground-truth-free argument about the **reading** step has the same shape, and the same gap. Defending what it reads on hidden layers, the foundational paper writes that “the scale, line separation, and script of the revealed characters are consistent with those observed on the fragment surfaces” [1]. The form is right — a verifiable region, an unverifiable one, and confidence transported by second-order statistics. It is left to the eye.

All four quantities are measurable without reading a letter: line spacing by autocorrelation of the row-density profile, character scale by connected components, ink coverage, and stroke width by the distance transform. We measured them on 190 published ink maps across four scrolls, and crossed them against the published ink contrast of one scroll’s 80 maps, split at its median.

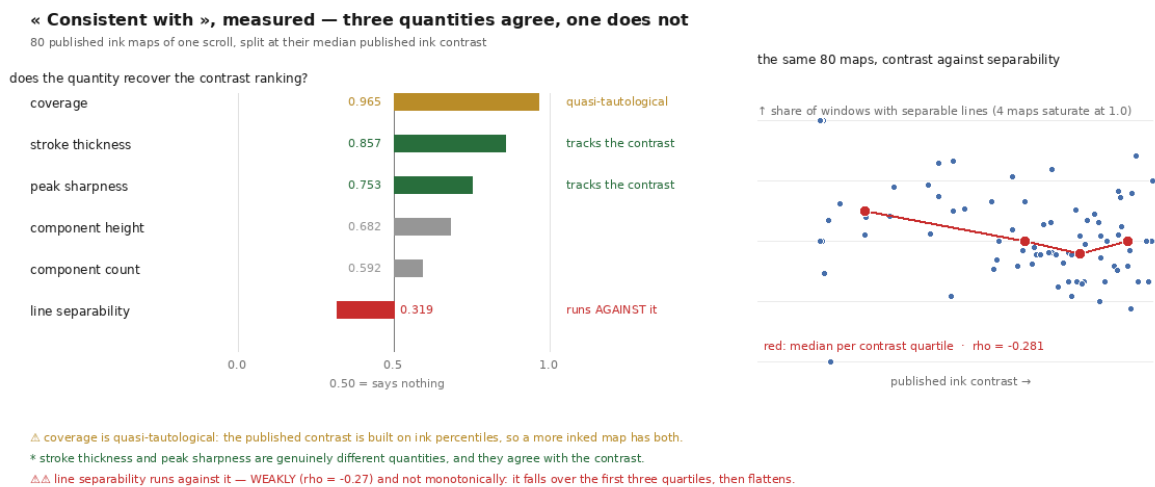


Figure 14: Each typographic quantity against the published ink contrast, as an AUC (0.50 = no association). Three recover the ranking. **Coverage is quasi-tautological** — the published contrast is built on ink percentiles — so the informative agreements are stroke width (0.857) and peak sharpness (0.753), which are genuinely different quantities obtained by a distance transform and an autocorrelation. Right: the same 80 maps behind the one quantity that runs the other way.

△△ And the three properties the sentence groups together do not go together. Line separability runs **against** the contrast (AUC 0.319, $\rho = -0.281$). The association is weak and not monotone — it falls over the first three quartiles and then flattens — and we do not present the obvious mechanism (heavier ink, lines closer to merging) as established. What matters here is the form of the finding: **an eye that judges a page “consistent” cannot see that two of its properties point in opposite directions. A number can.**

And this instrument has the same shape of limit as the first. A prediction that fails the typographic test is certainly not text; one that passes may be a periodic artefact — necessary, never sufficient, exactly as the absence of self-intersections is for a traced surface. Its periodicity floor is derived rather than chosen: white-noise autocorrelation over k lags of an n -point profile peaks near $\sqrt{2 \ln k} / \sqrt{n}$, so the floor rises by itself on a smaller map.

6.4 The control the field does not have

The same paper reports a false-positive rate for its ink detector [1], measured on images that contain ink all around. It never measures what the detector returns on a substrate **known** to carry none. The ideal control was in the scan — the paper sheet the fragments are mounted on, imaged in the same session at the same voxel — and the preprocessing removes it: “These are removed manually.”

A traced surface with a high convergence exponent is a better control than a paper backing, and it costs nothing extra. A surface with $\alpha \approx 1$ has a **geometric proof** that no papyrus face is within reach: its distance to matter follows the render window. Any ink reported there is a false positive by construction. We ran the published ink model on such a surface and, as a positive control, on the official segment of the **same scroll** — same volume, same voxel, same model, same region size, same stride. The only thing that differs is whether there is a sheet under the surface.

Two different volumes, one single answer

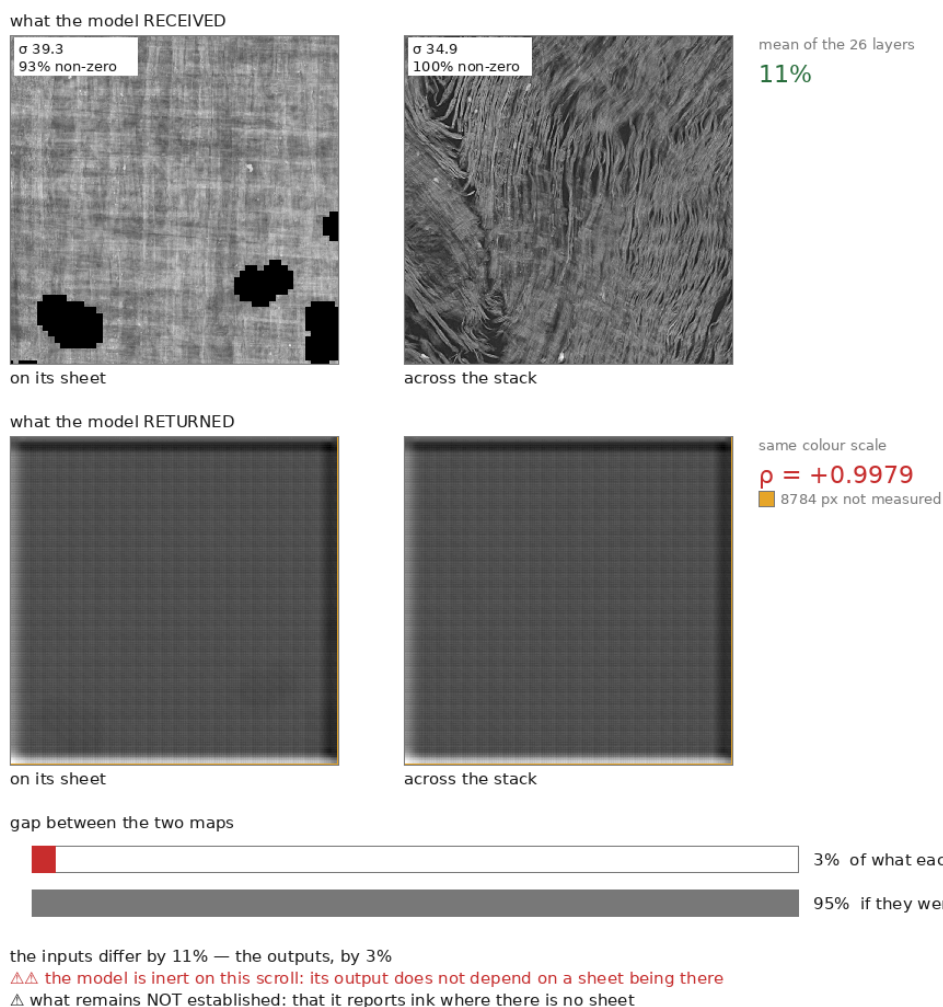


Figure 15: Top: what the model received — a woven papyrus face against a slice cut across the stack, two textures no eye would confuse, differing by 11.0 % in dispersion. Middle: what it returned, on a **shared** colour scale. Bottom: the gap between the two outputs against the gap two unrelated maps would show.

The intended claim is out of reach, and that comes first. On this scroll the detector returns a constant: $\sigma = 0.0129$, which is 1.7 % of the 0.7712 it returns where it reaches an AUC of 0.925. There is no working detector to control, so “it reports ink where there is no sheet” cannot be tested here — it reports none anywhere. Our instrument refuses to print a verdict below a tenth of the working dispersion, because two flat maps otherwise give a dispersion ratio near 1, which would read as “the control fools the detector” when it means “the detector is off on both sides”.

A narrower claim is established. A ratio of dispersions cannot separate **two maps of equal amplitude** from **the same map**, and that distinction is the one left: a detector returning two different maps is responding to its inputs, however weakly. Pixel to pixel, the two predictions correlate at $\rho = +0.9979$, and

their median difference is 2.9 % of what each map itself varies by — against 95.4 % for two unrelated maps of the same dispersion, a value derived rather than chosen.²

The boring explanation is excluded by measurement rather than by argument: both input windows are full and different — 93.4 % and 100 % non-zero, dispersions 39.26 against 34.95. The model receives two clearly distinct volumes and returns the same map.

The generalisable part is the design. A negative control for an ink detector does not require a special acquisition or a substrate known to be blank. It requires a surface whose **geometry** rules out the thing being detected, and any pipeline that can trace surfaces can produce one deliberately.

It also yields a cheap entry condition for a question this field has not answered: **does repairing a trace improve what is read from it?** Answering it requires a faulty trace, its repaired version, and the same downstream applied to both. On this scroll the downstream cannot separate them — a difference far larger than any repair leaves the output unchanged. Before spending compute on such an experiment, one can measure whether the detector responds on the target scroll at all: one inference on one window, and compare its dispersion to what the model returns where it is known to work.

Applied across the prize set, that condition is restrictive. Crossing the scrolls we can trace with those whose published ink maps carry the typographic statistics of a written page (Section 6.3) gives an **empty** intersection: 13 against 3, disjoint. The experiment is therefore not mountable on the prize set as it stands — not because repair cannot be measured, but because the scrolls where it could be read are not the ones that have been traced. Naming that before running the experiment is cheaper than discovering it after.

6.5 What a good measurement looked like here

Three properties recur in the instruments above and are worth naming.

Compare a thing to itself the convergence test changes a rendering parameter and reads the response. No reference surface is needed, which is precisely what a sealed scroll denies.

Prefer a dimensionless quantity α needs no threshold in microns, so it transfers between scrolls, resolutions and inter-sheet spacings without recalibration.

Make the refusal explicit several of our tools decline to answer rather than guess — on fewer than two windows, on windows too close together, on a scroll with a single run, on a pair where both members hit the budget. Each refusal was added after an instance where guessing produced a plausible and wrong number.

7 Limitations

1. **α has a resolution of about ± 0.2 on two windows** (Section 3.4). Counts of verdicts near the boundary are counts of coin flips, and we treat them as such.
2. **α is a median and hides a minority.** A surface can converge while a fraction of its windows has no sheet within reach. We report that fraction alongside, and the two measurements do sometimes disagree.
3. **The budget experiment covers two scrolls**, not the seven that plateau. It establishes that truncation explains the low dispersion where it was measured, not that every plateaued scroll behaves so.
4. **The remote triage does not replicate** outside `Scroll 1` (Section 4), and we do not know why. One of three negative corpora is explained by a floor effect; two are not.
5. **We do not know why a wrap fails.** Across four campaigns and forty wraps, the best predictor of a wrap’s α is its ordinal position in the chain ($\rho = +0.53$, $p = 0.001$) — better than erosion, arc length, area or any single-render statistic we tried. Depth is a clock, and we cannot yet separate it from a cause.
6. **The relative criterion is proposed, not built** (Section 5.4). We show that an absolute threshold on a profile criterion compares settings; we do not show that reading the same criterion as an exponent

²For two independent maps of dispersion σ , the difference has dispersion $\sigma\sqrt{2}$, so the median of its absolute value is $0.6745\sigma\sqrt{2} = 0.9539\sigma$.

works. It needs traces uncensored at both depths — 92 of our series qualify — so what is missing is the definition of the criterion, not the material to build it on.

7. **$\alpha \approx 1$ has two causes and does not separate them** (Section 3.5). A flat profile yields $\alpha \approx 1$ by identity. Both causes condemn the trace, and no converging verdict is affected, but 24 of 111 series in our tree cannot be said to be **across the stack** rather than **unmeasured**.
8. **The negative control establishes the narrow claim only** (Section 6.4). On the one scroll where we have both a sheet-following and a stack-crossing surface, the ink model returns a constant, so “it reports ink where there is no sheet” is untested. What is established — that its output does not depend on a sheet being present — rests on one scroll and one model.
9. **A render’s cost is dominated by a setting nobody had set, and by a wall nobody had measured.** Our renderer’s Zarr chunk cache defaults to 16 GB; none of the 28 call sites in our tree set it. On small surfaces this is invisible — the cache is allocated lazily and never fills. On a 3.66 cm² surface it fills, and together with the output buffers the process held **28.2 GB** on a 32 GB machine: 274 MB free, 3.4 GB swapped, and **23.7 % of one core** while 22 sat idle and the process carried 59 threads. It was not computing; it was waiting on memory. Capping the cache at 1 GB is measured to be both the smallest peak and the fastest median over 15 runs (peak **1.81 GB** against **4.53 GB** at the default), with all 15 outputs byte-identical — which is what licenses changing the setting at all. The wall behind it is harder. Buffers scale as area times depth: 41 layers of that surface need 9.7 GB, 161 need **38.2 GB**, and no cache setting moves that. The deep window of our own convergence test is therefore not slow on this machine, it is **impossible** — a fact no amount of patience discovers, and one that our published cost caveat (Section 5.2) priced in minutes when it should also have priced in gibibytes.
10. **Nothing here reads text.** The measurements judge geometry. A surface that passes every test in this paper may still carry no ink.

8 Conclusion

Segmentation quality can be measured on a sealed scroll, without ground truth, without a physical threshold and without a scale, by comparing a surface to itself under a change of rendering depth. Doing so on 13 prize scrolls shows that the reference tracer is a draw rather than a function, that the stability and cleanliness usually read as signs of a good trace are artefacts of the generation budget, that the render window imposes a second ceiling of the same kind, and that the published segmentation of a prize scroll samples sheets rather than tiling one. Each of the four was found by repeating a measurement that the field currently performs once. The same construction that judges a surface also supplies what the field’s reading step lacks: a surface whose geometry rules out a papyrus face is a negative control that costs one extra render.

Reproducibility. Every figure in this paper is produced by a script from a versioned result file, and every number quoted in the text is recomputed from that file and searched literally in the source of the paper by an automated guard. The rule the guard enforces is the one we would offer as the paper’s methodological summary: **a published number whose computation is not in the tree is not a result, it is an anecdote.**

References

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- [2] G. Angelotti *et al.*, “Complete virtual unwrapping and reading of a rolled Herculaneum papyrus,” *arXiv preprint arXiv:2606.29085*, 2026.
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- [4] G. Angelotti, F. Nicolardi, P. Henderson, and W. B. Seales, “Ink Detection from Surface Topography of the Herculaneum Papyri,” *Scientific Reports*, 2026, doi: 10.1038/s41598-026-58467-1.